Chapter XIII

Monge Closing a Circle

About three hundred years after Piero della Francesca presented his perspective construction based on the plan and elevation technique, this perspective method enjoyed a veritable renaissance as part of descriptive geometry. This brief chapter deals with the emergence of descriptive geometry, how the discipline shares one of its basic idea with Lambert's perspective geometry, and how perspective is treated in descriptive geometry.

XIII.1 Monge and Descriptive Geometry

Creation of Descriptive Geometry

The name of the French mathematician Gaspard Monge (1746–1818) has cropped up earlier in connection with a description of the further development of applying plans and elevations to make graphical representations. In the late 1760s, while teaching at the *École royale du génie* (The royal school of engineering) at Mézières, Monge began to combine the old plan and elevation technique with methods that would later become part of differential geometry. Gradually he developed a new discipline that he called descriptive geometry.[1]

During the French Revolution and the Empire, Monge was given important responsibilities, political as well as administrative. He was, for instance, very influential in renovating the Parisian system of higher education and in deciding the curricula at the *École normale* and the *École polytechnique*. At both these schools he made descriptive geometry an essential part of the syllabus. At the *École normale* he held thirteen lectures on descriptive geometry during 1795, and had his lecture notes duplicated. Under the editorship of Monge's assistant, Jean Nicolas Pierre Hachette, the first nine lectures appeared in 1799 as *Géométrie descriptive* – the last four lectures were not included, as Monge thought they needed to be refined (Taton^{S} 1951_3, 41–42).

[1]For a brief introduction to the history of descriptive geometry, see Andersen & Grattan-Guinness^{S} 1994. For thorough analyses of Monge's work, see Taton^{S} 1951_3 and Sakarovitch^{S} 1998.

Hachette's edition was reissued several times, and from 1820 it was enlarged with three additional lectures dealing with shadows, aerial perspective,[2] and linear perspective in an edition supervised by Monge's student Barnabé Brisson.

Monge's Descriptive Geometry

In innovating the plan and elevation technique, Monge rotated the half-plane of the elevation, *LMPQ* (figure XIII.1), down into the half-plane of the plan, *LMNO*, around their line of intersection, *LM*, and looked at the geometry in the plane consisting of the plan and the turned-down elevation. I call this plane the *descriptive plane* – Monge seemingly had no special term for it.

In the descriptive plane a point in space, say A, is represented by the two points a and a'' lying on a line perpendicular to the line LM. A line segment, say AB, is consequently represented by the pair of line segments ab and $a''\,b''$ connecting the representations of the two end points (if these points have either a common plan or a common elevation, one of the line segments degenerates into a point).

Monge described a number of fundamental connections between elements in three-dimensional space and their representations in the descriptive plane, such as the relationship between the length of a line segment and the lengths of the two line segments representing it.

More important for the present story, however, is the fact that Monge also showed how a number of standard Euclidean constructions can be performed directly in the descriptive plane. His very first example concerns the problem of how to construct the line through a given point that is parallel to a given line – where the latter does not contain the given point (Monge 1820, §14).

It is interesting to compare Monge's programme with Lambert's. Both applied a plane to represent the three-dimensional space – Lambert working with the perspective plane and Monge with the descriptive plane. Both mathematicians also became engaged in solving the problem of how to perform Euclidean constructions directly in their planes of representation. As Lambert's main work on perspective was published in 1759 – and actually in French – the possibility exists that Monge was inspired by Lambert's idea of working directly with the geometry in a representation. As noted in chapter XII, however, I have found no indications that Monge was familiar with Lambert's ideas or modelled his work on them.

---

[2]*Perspective aérienne*, by which Monge meant the science of light and shadow (Sakarovitch[S] 1998, 211).

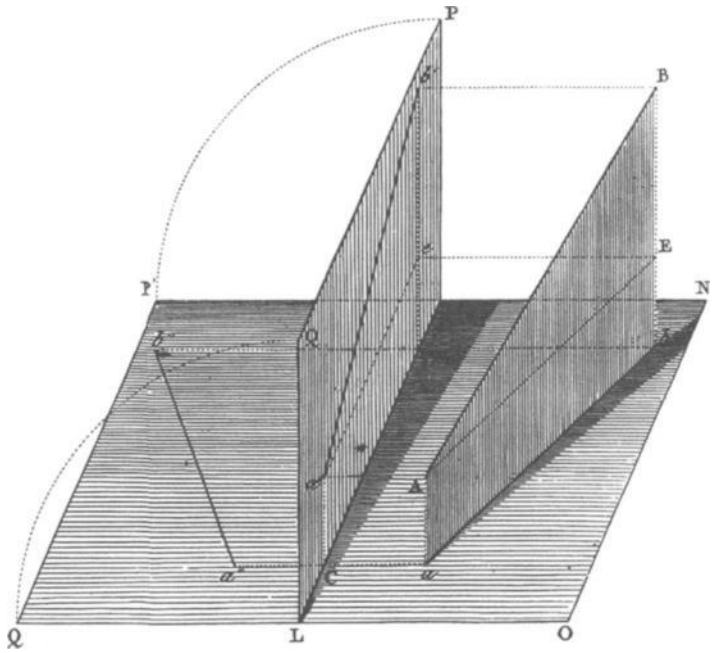

FIGURE XIII.1. A line segment projected upon the descriptive plane. Monge 1820, figure 2.

XIII.2 Monge and Linear Perspective

Monge's Presentation of Perspective

Monge's interest in linear perspective dates back to at least 1768, when he wrote a small treatise on the subject based on a plan and elevation construction (Sakarovitch[S] 1998, 231). Twenty-seven years later, while lecturing at the *École normale*, he took up perspective as a way to apply his descriptive geometry. His approach to perspective seems to be influenced more by the early writers than by authors closer to his own time. Monge thus introduced the idea of the picture plane as a pane of transparent glass, and he described a perspective image as a section in a visual pyramid (Monge 1820, §136).

Not only did Monge use a pyramid when introducing perspective, he also dealt with it in an example of throwing an object into perspective (figure XIII.2) – which is, in fact, his only example. Like many of his predecessors, he applied a plan and an elevation of the pyramid and the picture plane to determine what corresponds to the coordinates of the image points of the pyramid.

Compared to Piero della Francesca's presentation of a plan and elevation method for constructing perspective images, Monge's example offers

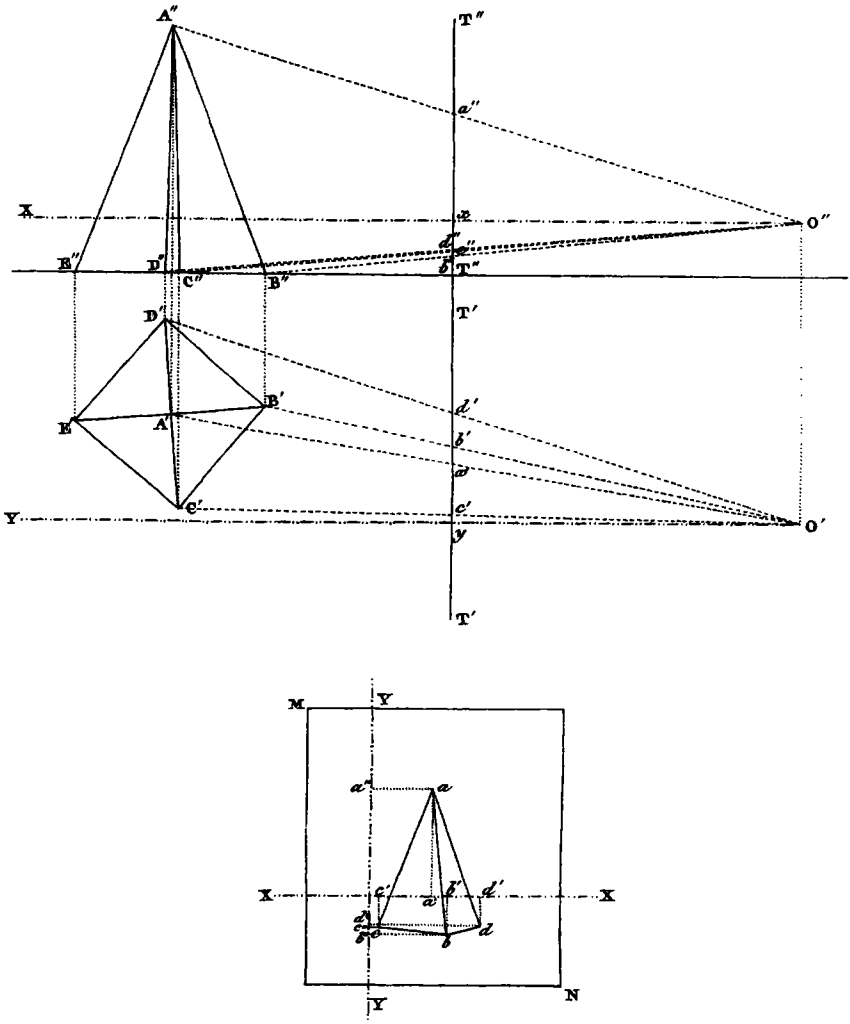

FIGURE XIII.2. Monge's example of throwing a pyramid into perspective. Monge 1820, figure 53.

absolutely nothing new. I therefore find it appropriate to say that Monge closed a circle when he incorporated perspective into his descriptive geometry.

In fairness, it should be noted that Monge did not stop his treatment of perspective after presenting his pyramid example. He touched upon the theory of perspective, for instance stating the result that the image of a line is generally a line, and presenting the vanishing point theorem. He did not, however, use his descriptive geometry to deduce these results, but obtained

them by looking at the three-dimensional situation, just as his predecessors had done (*ibid.*, §§136–137).

Besides his few theorems, Monge included a number of observations, asserting, for instance, that it was a good idea to determine the visible part of an object before throwing it into perspective. For objects bounded by planes, he found this could be done easily, whereas objects having curved surfaces could present complicated problems. For solving the latter type of problem, he referred readers to his treatment of shadows (*ibid.*, §136). He also held the opinion that students could benefit from their knowledge of descriptive geometry when the picture plane is a curved surface, or when they had to construct cylindrical mirror anamorphoses (*ibid.*, §138). Finally, Monge dealt with the problem of transforming a perspective composition to another picture plane than the one for which it was originally made, while keeping the same eye point.

Monge's Influence on Teaching Perspective

We know that, largely thanks to Monge's efforts, descriptive geometry became a topic that was taught at many institutions for more than a hundred years. However, no investigation has been made into how common it was to include perspective in courses on descriptive geometry – and it is outside the scope of this book to undertake such a study. I am therefore unable to say what the return to a plan and elevation approach to perspective meant to the life of perspective geometry, which was partly introduced by Taylor and fully developed by Lambert. It seems fairly safe, however, to conjecture that descriptive geometry did not in any way contribute to the mathematical understanding of perspective.

Chapter XIV

Summing Up

XIV.1 Opening Remarks

After all these pages on the history of perspective, the time has come to ask: What have we learned? In giving partial answers I return to the key issues and questions presented in the introduction. For structural reasons I have changed the order and also reformulated some of the questions. Thus, in the next pages I sum up my findings on the following points:

- geographical differences
- innovations in the mathematical theory of perspective
- the interplay between perspective and other geometrical disciplines
- the status of the theory of perspective
- practitioners' appreciation of the theory of perspective
- communication between theorists and practitioners
- the usefulness of the theory of perspective

Finally, I ponder the driving forces behind the theory of perspective.

Before addressing the points mentioned above, let me make a few general comments upon my source material. We have seen a few instances of books simply being copied[1] – and I do not guarantee that there are not other examples of this. We have also met highly original publications that inspired other authors as concerns their content, but not their style. The case of Guidobaldo del Monte and his influence on the works of Simon Stevin (1605), François Aguilon (1613), and Samuel Marolois (1614) is the most spectacular example, but the British literature inspired by the work of Brook Taylor (1719) is also noteworthy. A large part of the literature lies between these two extremes, by which I mean that it is rather derivative without containing passages copied verbatim. I assume that an investigation of textbooks on other subjects would lead to a very similar result.

[1]Thus, in 1505 Gregor Reisch copied parts of Viator's *De artificiali perspectiva*, and in 1509 Jörg Glockendon copied the entire book (page 166). Similarly, in 1790 Robert Bradberry published a book in which most of the content was taken from Benjamin Martin *s.a.* and the rest from William Hooper 1774 (page 591).

My last opening remark concerns women; I have not been able to find a single female author. My impression is that, particularly in the eighteenth century, many women took lessons in painting, and may well have been taught some rudimentary perspective techniques. Apparently, however, none of these women became so keen as to study perspective in such depth that they themselves wished to publish on the subject – although some women did write on science topics during this period.[2] The only woman mentioned in this book in connection with perspective is Mary Somerville – and she falls outside the period considered. She was taught perspective, and in this way she came to learn about Euclid's *Elements* (page 598), but rather than becoming interested in the mathematical aspect of perspective she turned her interest to other mathematical sciences.

XIV.2 Local Approaches to Perspective

Knowledge about perspective did cross boarders, but I nevertheless find that by treating contemporary authors from the same area together I have been able to draw a picture of substantial regional diversities. Within each country there were different approaches to perspective, but we can still distinguish some general trends concerning the interest in the subject and the seriousness with which perspective constructions were treated in various periods. Occasionally repeating some of my earlier conclusions, I will, in the following paragraphs, survey the main lines in the geographically diverse developments.

The Italian Development

Emerging among a circle of Florentine artists, perspective became a much-desired form of representation in painting during the fifteenth and sixteenth centuries. In the *quattrocento* two important texts on perspective appeared, written, respectively, by Leon Battista Alberti and Piero della Francesca. Both of these authors were clearly inspired by the painters and architects with whom they were in close contact – actually belonging themselves to these groups. In the *cinquecento*, a considerable number of publications on perspective appeared in Italy, and in several of these the authors showed a keen interest in coming to understand the geometry behind perspective, as Piero had done earlier. This development led to Guidobaldo's revolutionary approach to perspective, published in 1600. Seen from a mathematical point of view, the Italian literature was dominating until that date.

In the seventeenth century, the interest of Italian patrons, painters, and mathematicians in perspective waned considerably, and at the same time the quality of the publications on the subject generally dropped. It is particularly

[2]To my knowledge, the first textbook on perspective written by a woman is Le Breton 1828.

remarkable that no Italian scholar followed up on Guidobaldo's ideas. Torricelli would have been more than capable of doing this, but he chose a practical approach, being of the opinion that the theory of perspective was too fatiguing for practitioners, and apparently taking no personal academic interest in the subject. As a result, the Italian tradition of writing on perspective was mainly kept alive by stage designers and painters who specialized in illusionistic paintings.

The French and Belgian Development

Just as perspective drawing and painting in Northern Europe was inspired by Italians but found its own expression, so did the literature on perspective. Viator's publications from the early sixteenth century were distinctly different from the Italian texts in style. As the century wore on, some new knowledge emerged in France, demonstrated in Jean Cousin's work from 1560, but probably from an earlier unknown source. Cousin also had a way of expressing himself that was not modelled upon Italian publications.

In the seventeenth century, perspective caught the attention of many French and a few Belgian scientists and mathematical practitioners. The first group presented the subject in general textbooks on mathematics and other sciences, while the second group produced separate volumes on perspective in which they took a new approach, introducing perspective scales and performing constructions directly in the picture plane. Later in the seventeenth century, the intense activity in the field of perspective declined. Some textbooks still appeared during the period up to 1800, but they did not show any innovation, nor did they take up the advanced ideas presented by Brook Taylor and Johann Heinrich Lambert.

The German Development

Clearly inspired by Italian sources, Albrecht Dürer published the first description in German of a perspective construction, or rather of two constructions. Dürer awakened an enthusiasm for perspective in a Nurembergian group of artisans, several of whom published on the topic. It is characteristic for the early German works on perspective that they are geometrically imprecise, for instance by not including descriptions of all the steps needed to construct the perspective image of a figure. These books also show a profound fascination with perspective instruments.

Just as the German authors were late in giving distinct presentations of perspective constructions, the mathematical theory of perspective also came late into the German literature. Eventually, at the end of the seventeenth century, it became common to include a section on perspective in German textbooks on mathematics. However, not until the 1750s did these presentations reach the level of exactness that had been common in general books on mathematics in France and the Netherlands for almost one and a half centuries.

The first comprehensive mathematical work on perspective written by a German was published in 1775 – and its author was Wenceslaus Johann Gustav Karsten.

The Dutch Development

In the seventeenth-century Northern Netherlands, the involvement in providing some geometrical background to perspective ran parallel with interests in creating and acquiring paintings with striking perspective compositions, as had been the case during the Renaissance in Italy. In fact, Simon Stevin, Samuel Marolois, and Frans van Schooten produced extremely good texts on the theory and practice of perspective during the golden era of Dutch painting. Both the academic and the practical interest in perspective stayed alive until the beginning of the eighteenth century, culminating with Willem 'sGravesande's work from 1711. Then came a long period without any Dutch publications on perspective. From 1765 the Dutch scene was dominated by the engraver Caspar Jacobszoon Philips, who was an extremely dedicated teacher of the practice of perspective.

The British Development

Generally on the Continent, with the notable exception of Germany, the interest in perspective dwindled during the last two thirds of the eighteenth century. Not so in Britain, however, where the writing on perspective actually flourished in this period.

As was the case in Italy and the Northern Netherlands, the development in Britain may be taken as an example of the interest in perspective painting and writing on perspective constructions running parallel – though in the opposite sense, so to speak: The British only became interested in perspective drawings very late, and the same holds true for British writing on perspective. On the other hand, the literature then developed quickly within a fairly short span of years, during which perspective became well understood and well presented in Britain. Contrary to what was happening on the Continent, one person dominated the literature in Britain, namely Brook Taylor. Actually, a considerable portion of the numerous British presentations of perspective appearing in the second half of the eighteenth century were directly related to Taylor's work.

XIV.3 Perspective and Pure Mathematics

Innovations in the Mathematical Theory of Perspective

Although perspectivists were mainly inspired by authors from their own geographical circles, some learned from other sources. In the development of the geometrical understanding of perspective at the highest level there was

clearly a continuous development in the sense that when an author took an important new theoretical step, he was, in almost every instance, building upon the most advanced theory of his day.

For the main development, I have documented the following trajectory of inspiration: An interest among Italian mathematicians in the latter half of the sixteenth century animated Guidobaldo to take up perspective. Through a gradual process, he came to realize that not only orthogonals and diagonals have vanishing points, and saw the importance of the main theorem of perspective. Guidobaldo's insights travelled from Italy to the Northern Netherlands and were quickly adopted by Stevin. Directly influenced by Stevin, and possibly also by van Schooten, 'sGravesande picked up the theory of perspective. He did not add any decisive new results to the theory, but he appreciated and exploited its potential more strongly than his predecessors, for instance by focussing upon a very elegant and simple visual ray construction. Then, once more, the development crossed a border or, to be more precise, a sea. Taylor was inspired by 'sGravesande's mathematical understanding and added quite a bit of his own thinking. In fact, Taylor provided perspective with a new mathematical life, among other things by introducing and applying the general concept of a vanishing line.

I have not been able to trace the sources for Lambert's mathematical knowledge of perspective, but we can be sure he did not reinvent the theory. His work was part of the continuous development of the theory of perspective, and he brought this theory as far as it could presumably be taken as an independent discipline. By this I mean that as for the specific question of how to project three-dimensional figures upon a plane surface, no important question seemed to have been left unanswered. Nor did there seem to be any way of carrying out perspective constructions more elegantly than by the methods advocated by Taylor and Lambert. As a consequence of this success, the theory of perspective became less attractive for mathematicians, to whom a field with no loose ends holds no appeal.

Interplay Between Perspective and Other Geometrical Disciplines

From Commandino onwards, several mathematicians remarked that a perspective projection is just one of several examples of a central projection, and not rarely perspective was presented alongside other projections. However, no general theory of central projection was developed that could be applied to obtain results concerning perspective constructions. In fact, almost all the fundamental results in the theory of perspective were obtained on the basis of knowledge that have been around for almost two thousand years, primarily the theory contained in Euclid's *Elements*. In connection with investigations of the perspective images of a circle, Apollonius's theory of conics was also applied. In other words, the contents of the perspectivists' mathematical toolbox consisted, first and foremost, of classical geometry.

Conversely, the theory of perspective did not offer much inspiration to other fields. Girard Desargues may have been influenced by his theoretical consideration on perspective when he introduced points at infinity – but other mathematicians introduced this concept as well, and did so independently of perspective. Similarly, cross ratios are to be found implicitly in Taylor's work, but it was not through the theory of perspective that mathematicians later realized that a cross ratio is an important concept in projective geometry, and that cross ratios are invariant under a central projection.

Lambert came farthest in relating perspective to other disciplines. The elegance of his application of the theory of perspective to obtain a result that was a forerunner of Pohlke's theorem for parallel projections is striking. Even more impressive is his involvement of the theory of perspective in attacking the question of when a traditional construction problem can be solved using a ruler alone. If Lambert had lived longer and had continued his investigation of ruler geometry, he might have anticipated more of Poncelet's work, but as matters stand, the intensification of the work on central projection and the creation of projective geometry during the nineteenth century happened independently of perspective geometry.

The Status of the Theory of Perspective

In early modern times a distinction was made between pure and mixed mathematics, and perspective was grouped in the latter category (for a discussion of perspective as a mixed science, see Peiffer[S] 2002). This is understandable, as the theory of perspective was created for the explicit purpose of providing a foundation for perspective constructions. In nature, however, theoretical problems on perspective are somewhat similar to classical pure geometrical problems, and could therefore potentially be seen as an interesting part of theoretical mathematics, but that did not happen. Christiaan Huygens found geometrical perspective too easy to be of real interest, and 'sGravesande claimed that the theory was "barren of great discoveries" (pages 319 and 360). In 1758 Jean Étienne Montucla expressed a similar view, characterizing perspective as "a subject in which the greatest difficulties do not go beyond the reach of a mediocre geometer".[3]

Comparing problems that were solved in perspective with those solved in classical geometry, I do not find that the perspective problems were significantly easier than the geometrical ones. Hence, the only way I can understand the attitude to perspective expressed by some seventeenth and eighteenth-century mathematicians is that to be really interesting in those days, a topic had to relate to the two new important mathematical disciplines: analytical geometry and calculus.

---

[3] *Un sujet dont les plus grandes difficultés ne sont pas au-dela de la portée d'un Géometre médiocre.* [Montucla[S] 1758, 636]

Regardless of how the mathematicians' sentiments on the theory of perspective should be explained, they had the ironic effect that the more advanced aspect of the theory – the part concerning vanishing lines and direct constructions – only became known to a select few. The reason for this was, as stressed in other chapters, that literature on perspective was in general dedicated to practitioners, who lacked sufficient mathematical education to understand what the mathematicians had written for them. If the theory of perspective had been held in higher regard within mathematics, more publications on perspective would presumably have been addressed to the mathematicians.

XIV.4 The Theory and Practice of Perspective

The Practitioners' Appreciation of the Theory of Perspective

As noted, there were three major steps in the development of the theory of perspective: Guidobaldo's introduction of the general notion of a vanishing point and the main theorem of perspective, Taylor's development of the generalized concept of a vanishing line, and Lambert's creation of an abstract perspective geometry.

The two last-mentioned innovations did not have any real impact on how perspective was presented in the practitioners' textbooks. The concept of vanishing lines was introduced in most of the Taylor-inspired English literature from the second half of the eighteenth century, but it was hardly ever applied, and outside the British isles it is all but impossible, to find a vanishing line mentioned in a textbook written by a practitioner. In the period examined, Lambert's perspective geometry was almost totally ignored by practitioners and certainly had no influence on the textbooks they composed.

As to vanishing points, it is rare to see practitioners writing about other vanishing points than the principal vanishing point and the distance points. Still, it is my impression that Guidobaldo's work was instrumental in making many practitioners aware of the concept of a vanishing point as a convergence point for the images of parallel lines, and a few may even have appreciated that a proof of this property existed. In short, among practitioners of perspective there was no great interest in or appreciation of the theory developed after Guidobaldo, although they did, directly or indirectly, apply the theory of vanishing points.

Communication Between Theorists and Practitioners

Throughout this book we have met numerous examples of authors who claimed that the state of communication between the theorists and practitioners of perspective was deplorable – and many more examples can be

found. The practitioners tended to blame the theorists for the lack of collaboration, and vice versa.

This communication gap is not particular to perspective, but also occurs in several other disciplines where mathematics is used. The problem seems to be that people applying a mathematical discipline are asking to have the contents of mathematics presented without too many mathematical terms, and that the mathematicians take it for granted that the basic elements of mathematics can be quickly understood by readers with no mathematical background.

In the case of perspective, a number of attempts were made to bridge the gap, most of which proved ineffective, whereas some of the practitioners who transmitted Taylor's theory in the second part of the eighteenth century to a general British audience were rather more successful.

The Usefulness of the Theory of Perspective

Having once understood the contents of Lambert's work on perspective, there is almost no end to what one can draw correctly in perspective: all polyhedra, several solids with curved surfaces, reflections, water jets, rainbows, a starry sky, and perspective pictures in a composition that has itself been thrown into perspective. Lambert also offered the solution for equipping an oblique ceiling with a *trompe l'œil*, or for making an oblique garden appear horizontal. The big question is, however, whether anyone felt the need to put any of this marvellous theory into practice. My feeling is that no one did. Most drawers planning to make a perspectival composition would presumably be well equipped with a three-dimensional coordinate system in the picture plane, already provided by Vredeman de Vries (figure V.74), supported by a basic understanding of the concept of a vanishing point – including the knowledge that not all vanishing points are situated on the horizon. Actually for his teaching of perspective three hundred years later Joseph Mallord William Turner used a diagram inspired by Vredeman de Vries (figure XIV.1). The elaborate use of the theory of perspective was reserved for those who took pleasure in getting the perspectival details correct, and they were, in all probability, very few.

For the sake of completeness, let me add that in the nineteenth century when photometric surveying was introduced, the theory of perspective, more precisely the part concerning inverse problems of perspective, became highly useful – at which point my story had already ended.

XIV.5 The Driving Forces Behind the Theory of Perspective

In this chapter I have stated that in the period I have studied, the theory of perspective did not contribute to the development of other branches of mathematics, that the discipline in general was not held in high regard by

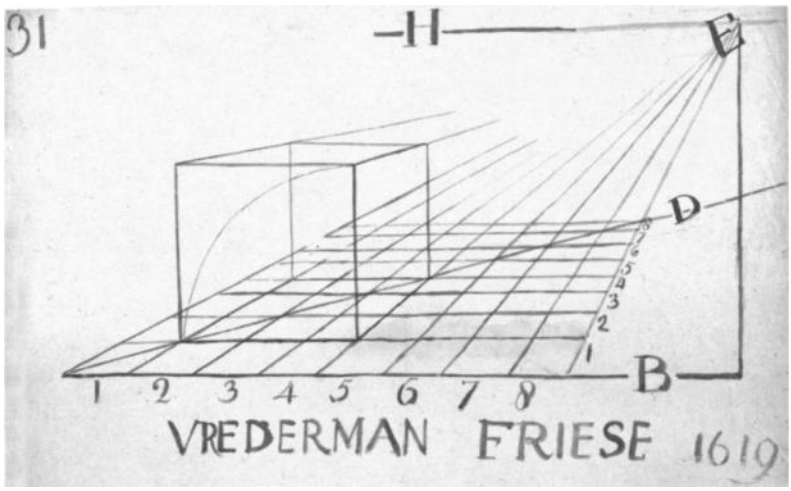

FIGURE XIV.1. Turner, Lecture Diagram 31: Perspective Method for a Cube by Vredeman de Vries, c. 1810–1827. Tate Britain.

mathematicians, and that only a few of its results were applied. When considering the collective implications of these views, one wonders what the point was of developing the theory of perspective any further than to the concept of a vanishing point.

This was not a question the main protagonists asked. The forces that drove them were their interest in and enthusiasm for the theoretical aspects of perspective, and their inclination to give in to the seduction of mathematics.

In their publications the theorists claimed that practitioners would benefit from their work, and some of them may have been so naive that they really believed this. However, they may also secretly have hoped that other mathematicians might share their feeling for the beauty of their most elegant results. To what extent this actually happened, we will never know, but we can be sure such a meeting of the minds took place at least once, when Johann Bernoulli expressed his admiration for 'sGravesande's mathematical abilities after reading his *Essai de perspective* (page 359).

A number of mathematical practitioners, especially among the Britons who propagated Taylor's perspective theory, also fell prey to the seductive attraction of the geometrical theory of perspective. Thus, summing up in the words of Thomas Malton: "Of all the Mathematical Sciences, the study of Perspective is perhaps the most entertaining" (T. Malton 1775, ii).